DOCUMENT 03

Visualization and Dashboard

Website pages, dashboard design, Tableau integration, and testing

Project: Olympic Sports Performance Analysis

Application: Python Flask web application | **Deployment:** Render

1. Purpose of Visualization

Data visualization converts summaries and comparisons into visual forms that are easier to understand. The project provides separate Dashboard and Story pages so that users can explore the intended Olympic analysis through a website.

2. Website Navigation

Page	Route	Purpose
Home	/	Introduces Olympic Sports Performance Analysis.
Dashboard	/dashboard	Prepared to embed an interactive Tableau dashboard.
Story	/story	Prepared to embed a Tableau story with a sequence of views.

3. Suggested Dashboard Components

- **Summary cards:** Total medals, participating countries, athletes, or sports, if these measures are available and consistently defined.
- **Time-series chart:** Participation or medal trends across Olympic years.
- **Country comparison:** A bar chart comparing medal totals for selected countries.
- **Sport-wise comparison:** A chart showing medal distribution by sport.
- **Medal composition:** A chart separating gold, silver, and bronze medals.
- **Filters:** Year, season, country, and sport filters where supported by the underlying data.

4. Recommended Chart Selection

Analytical question	Suitable visual
Change over time	Line chart
Compare countries or sports	Sorted bar chart

Analytical question	Suitable visual
Medal category share	Stacked bar or simple composition chart
Geographical comparison	Map, if country fields are standardised
Detailed records	Table

5. Tableau Integration in Flask

The dashboard and story templates load Tableau's Embedding API JavaScript module and use a **tableau-viz** element to display a published view. The Flask routes render the corresponding HTML templates.

Important implementation status: In the supplied ZIP, the Tableau src values are example placeholders containing REPLACE_WITH_WORKBOOK and REPLACE_WITH_VIEW (and equivalent story placeholders). Replace them with the actual published Tableau URLs and confirm that the workbook permissions allow the intended viewers to access them.

6. Design and Usability Guidelines

- Use clear titles, labels, units, and legends.
- Keep colours consistent for medal categories and avoid unnecessary decoration.
- Make filters easy to find and explain what they change.
- Check the dashboard on desktop and mobile screen sizes.
- Provide a short insight or takeaway near important charts.

7. Testing Checklist

- Open /, /dashboard, and /story and confirm that each page loads.
- Confirm that published Tableau views load without permission or URL errors.
- Test filters and interactions inside Tableau.
- Check layout, readability, and responsive behaviour.
- Verify that chart totals match the cleaned source data.

Documentation note: This document describes the project files supplied for review. The supplied ZIP does not include the source Olympic dataset or completed Tableau workbook. Therefore, no medal totals, rankings, chart values, or statistical findings are invented here. Update the relevant sections after connecting the final dataset and publishing the Tableau views.